Task 1: Order Flow Imbalances (OFI)

Objective

Evaluate the candidate's understanding of price impact models, Order Flow Imbalance (OFI) construction, and the ability to extend OFI concepts across different order book granularities and cross-asset relationships. This skill is foundational to our research efforts.

Resources Provided

- Dataset: first_25000_rows.csv
- Research Paper: *"Cross-Impact of Order Flow Imbalance in Equity Markets"* (PDF)

Task Instructions

You must construct the following OFI features (your output should produce **one value per chosen timestamp**):

- **Best-Level OFI**
- **Multi-Level OFI**
- **Integrated OFI**
- **Cross-Asset OFI**

In addition, answer the following conceptual questions:

1. What's the motivation behind measuring OFI at multiple depth levels of the order book?
2. Why do the authors use Lasso regression rather than OLS for estimating cross-impact?
3. Why is OFI considered a better predictor of short-term returns than trade volume?

Submission Requirements

- **Code:** Upload your OFI feature construction code (clean and modular) to a **GitHub repository** and share the link.

- **Written Answers:** Submit a **LaTeX-generated PDF** that answers the conceptual questions clearly and professionally.

Task 2: Smart Order Routing (SOR) – Reading Preparation

Objective

Prepare the candidate to critically understand a foundational paper on smart order routing optimization ahead of the interview. This material will form the basis of technical discussion during the hiring process.

Resources Provided

- Research Paper: *"Optimal Order Placement in Limit Order Markets"* by Cont and Kukanov (PDF)

Task Instructions

- Carefully read and study the paper. Focus on understanding:
  - The formulation of the optimization problem.
  - How market orders and limit orders are allocated.
  - The role of sampling and stochastic optimization methods.
  - How market microstructure inputs (e.g., queue size, order flow) influence routing decisions.
- You will not be required to submit a written assignment for this task.
- However, **you will be asked technical questions based on this paper during the interview**, so please ensure you critically understand the material.

Final Submission Checklist

Deliverable	Format
Task 1 Code (OFI feature construction)	GitHub repository link
Task 1 Conceptual Answers	LaTeX PDF

Submission Deadline

All submissions are expected to be completed and sent within **48 hours** after the task is assigned (consider this is the most up to date deadline).
Please prioritize clean, readable code and clear, well-structured LaTeX writeups.

Would you like me to also create a **small message** you could paste when you send this task to candidates? (Polite and clear tone.) It'll make your outreach much smoother.
Let me know!